Gomory's Cutting Plane Method

Gomory's method refines the LP relaxation by adding cuts that "trim away" the fractional solutions, pushing the optimum towards integrality without branching.

Algorithm: Gomory's Cutting Plane Method:

1. **Solve the LP relaxation** of the integer programming problem.

2. **Check integrality**:
   - If all variables are integers, the solution is optimal.
   - If not, proceed to Step 3.

3. **Select a row** in the simplex tableau corresponding to a basic variable with a **fractional value (maximum)**.

4. **Construct a Gomory cut**: From the chosen row, write a new linear inequality that removes the fractional solution but keeps all integer feasible points.

5. **Add the cut** as a new constraint to the system.

6. **Re-solve** the new LP (using simplex or dual simplex).

7. **Repeat** Steps 2–6 until an integer solution is obtained.

Ex: Solve the IPP by Gomory's cutting plane method:

$$Max\ Z = 7x_1 + 10x_2$$

Subject to: $-x_1 + 3x_2 \leq 6,\ 7x_1 + x_2 \leq 35, x_1 \geq 0, x_2 \geq 0.$

$x_1,\ x_2$ are positive and integers.

Sol: After solving the given LPP by Simplex method, The optimal table is given by:

		$C_j \rightarrow$	7	10	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
10	x_2	$7/2$	0	1	$7/22$	$1/22$
7	x_1	$9/2$	1	0	$-1/22$	$3/22$
$Z_j - C_j \longrightarrow$			0	0	$63/22$	$31/22$

Here both the variables x_1, x_2 not integer with same fractional values. So, choose any row to construct the Gomory's constraint.

We choose the first row: $1.x_2 + \frac{7}{22}s_1 + \frac{1}{22}s_2 = \frac{7}{2} \quad \Longrightarrow \quad x_2 + \left(0 + \frac{7}{22}\right)s_1 + \left(0 + \frac{1}{22}\right)s_2 = 3 + \frac{1}{2}$

$$\Longrightarrow \quad x_2 - 3 = \frac{1}{2} - \frac{7}{22}s_1 - \frac{1}{22}s_2$$

Gomory's Constraint: $\frac{1}{2} - \frac{7}{22}s_1 - \frac{1}{22}s_2 \leq 0$

We introduce a slack variable s_3 in the Gomory's constraint: $-\frac{7}{22}s_1 - \frac{1}{22}s_2 + s_3 = -\frac{1}{2}$

		$C_j \rightarrow$	7	10	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3
10	x_2	$7/2$	0	1	$7/22$	$1/22$	0
7	x_1	$9/2$	1	0	$-1/22$	$3/22$	0
0	s_3	$-1/2 \longrightarrow$	0	0	$-7/22$	$-1/22$	1
$Z_j - C_j \rightarrow$		$Z = 0$	0	0	$63/22$	$31/22$	0
$\theta_{ij} \longrightarrow$			–	–	$\|-9\| \uparrow$	$\|-31\|$	–
10	x_2	3	0	1	0	0	1
7	x_1	**$32/7$**	1	0	0	$1/7$	$-1/7$
0	s_1	$11/7$	0	0	1	$1/7$	$-22/7$
$Z_j - C_j \rightarrow$			0	0	0	1	9

Here x_1 is not integer, so we introduce a Gomory's constraint corresponding to the x_1 row.

$$x_1 + \frac{1}{7}s_2 - \frac{1}{7}s_3 = \frac{32}{7} \quad \Rightarrow \quad x_1 + \left(0 + \frac{1}{7}\right)s_2 + \left(-1 + \frac{6}{7}\right)s_3 = 4 + \frac{4}{7}$$

Gomory's Constraint: $\Rightarrow \quad -\frac{s_2}{7} - \frac{6}{7}s_3 \le -\frac{4}{7}$

We introduce a slack variable s_4 in the Gomory's constraint: $-\frac{s_2}{7} - \frac{6}{7}s_3 + s_4 = -\frac{4}{7}.$

		$C_j \rightarrow$	7	10	0	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3	s_4
10	x_2	3	0	1	0	0	1	0
7	x_1	32/7	1	0	0	1/7	−1/7	0
0	s_1	11/7	0	0	1	1/7	−22/7	0
0	s_4	**−4/7 →**	0	0	0	− 1/7	−6/7	1
$Z_j - C_j \rightarrow$			0	0	0	1	9	0
$\theta_{ij} \longrightarrow$			–	–	–	$\lvert -7 \rvert \uparrow$	$\left\lvert -\frac{21}{2} \right\rvert$	–
10	x_2	3	1	0	0	0	1	0
7	x_1	4	1	0	0	0	− 1	1
0	s_1	1	0	0	1	0	− 4	1
0	s_2	4	0	0	0	1	6	− 7
$Z_j - C_j \rightarrow$		58	0	0	0	0	3	7

The optimal solution of the given LPP is

$$\boldsymbol{Max\ Z = 58, \qquad x_1 = 4, \qquad x_2 = 3.}$$

Ex: Solve the IPP by Gomory's cutting plane method:

$$Max\ Z = x_1 + 2x_2$$

Subject to: $x_1 + x_2 \leq 7,\quad 2x_1 \leq 11,\quad 2x_2 \leq 7,\quad x_1 \geq 0, x_2 \geq 0.$

$x_1,\ x_2$ are positive and integers.

		$C_j \rightarrow$	1	2	0	0	0	Min Ratio
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3	$\frac{b}{a_{ij}(>0)}$
0	s_1	7	1	1	1	0	0	7
0	s_2	11	2	0	0	1	0	-
0	s_3	7	0	2	0	0	1	$5 \rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	-1	$-2 \uparrow$	0	0	0	
0	s_1	10	1	0	1	0	$-1/2$	10
0	s_2	3	2	0	0	1	0	$3/2 \rightarrow$
2	x_2	$7/2$	0	1	0	0	$1/2$	$-$
$Z_j - C_j \rightarrow$		$Z = 9$	$-1 \uparrow$	0	0	0	1	
1	x_1	$7/2$	1	0	1	0	$-1/2$	
0	s_2	4	0	0	-2	1	1	
2	x_2	$7/2$	0	1	0	0	$1/2$	
$Z_j - C_j \rightarrow$		$Z = 11$	0	0	1	0	$1/2$	

Here we construct the Gomory's Constraint with respect to the x_2 row.

$$x_2 + \frac{1}{2}s_3 = \frac{7}{2} \quad \Longrightarrow \quad x_2 + \left(0 + \frac{1}{2}\right)s_3 = 3 + \frac{1}{2}.$$

Gomory's Constraint: $\frac{1}{2} - \frac{1}{2}s_3 \le 0 \Rightarrow -\frac{1}{2}s_3 \le -\frac{1}{2}$ We introduced a slack variable s_4, we get: $-\frac{1}{2}s_3 + s_4 = -\frac{1}{2}$.

		$C_j \rightarrow$	1	2	0	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3	s_4
1	x_1	7/2	1	0	1	0	$-1/2$	0
0	s_2	4	0	0	-2	1	1	0
2	x_2	7/2	0	1	0	0	1/2	0
0	s_4	**-1/2**$\rightarrow$	0	0	0	0	$-1/2$	1
$Z_j - C_j \rightarrow$		$Z = 11$	0	0	1	0	1/2	0
$\theta_{ij} \longrightarrow$			–	–	–	–	$\lvert -1 \rvert \uparrow$	–
1	x_1	4	1	0	1	0	0	-1
0	s_2	3	0	0	-2	1	0	2
2	x_2	3	0	1	0	0	0	1
0	s_3	1	0	0	0	0	1	-2
			0	0	1	0	0	1

$\boldsymbol{x_1 = 4, x_2 = 3,}$
$\boldsymbol{Z = 10}$